ABHISHEK KAR

Buffalo, NY | 612-450-8439 | karabhishek1222@gmail.com | Abhishek LinkedIn | Abhishek Portfolio

SUMMARY

Mechanical Design Engineer with an MS in Mechanical Engineering and 5 years of experience who owns 3D models, component and assembly drawings, and engineering specifications end to end, from manufacturing, quality, and customer requirements through release.

- **Precision Design:** Owned design of components and precision assemblies for 8+ electromechanical equipment programs in SolidWorks and NX, cutting rework by 31%.
- **Machine & Tooling Design:** Drive design of mechanisms, frames, brackets, mounts, fixtures, and tooling to customer specifications with minimal supervision.
- **Drawings & PLM:** Own complete drawing packages aligned with GD&T (ASME Y14.5) and drafting standards, along with BOMs and ECOs in Teamcenter and SAP-PLM.
- **Root Cause & Reliability:** Led root-cause analysis and corrective actions across 20+ production systems using Lean/OEE and Minitab, cutting incidents by 15%, reducing downtime by 6%, and saving about $50K per year.

SKILLS

CAD & Design: Siemens NX (transferable to Solid Edge), Creo, SolidWorks, CATIA V5/V6, Autodesk Inventor (Certified), AutoCAD Mechanical, mechanisms, fasteners, bearings, springs, seals, motors, sensors, actuators, enclosures, structural and component design

GD&T, MBD & PLM: GD&T (ASME Y14.5) with position and profile controls, datum structures, MBD (3D PMI, ASME Y14.41), component and assembly drawings, Teamcenter, Windchill, SAP-PLM, SolidWorks EPDM/PDM, BOMs, ECO/ECN

Analysis & Simulation: ANSYS FEA (structural, thermal, modal, nonlinear contact), Abaqus, fatigue, vibration, ANSYS Icepak, CFD (ANSYS Fluent), heat transfer, worst-case, RSS, and Monte Carlo tolerance stack-ups, topology optimization, parametric optimization, hand calculations

Automation & Programming: Python, MATLAB, NX Open, SolidWorks API, simulation automation, AI-assisted CAD

DFM/DFA & Materials: injection molding, die casting, sheet metal, CNC machining, welding, 3D printing, material selection (metals, polymers, coatings), first article inspection, supplier collaboration

Validation & Quality: EVT/DVT/PVT, DOE, failure analysis, reliability testing, root cause/CAPA, DFMEA, PFMEA, Lean/OEE, Minitab, Microsoft Office (Excel, Word, PowerPoint, Outlook)

WORK EXPERIENCE

Advanced Energy / L&T Technology Services — **Jun 2025 - Present**
Senior Mechanical Engineer — *New York, USA*

- Led **mechanical design** in **SolidWorks** and **NX** for **components**, **structural brackets**, **mounts**, and **precision assemblies** across **8+ electromechanical equipment programs**, reducing **rework by 31%**.
- Delivered complete **drawing packages** for **20+ released parts** with **GD&T drawings** and **188 part specifications**, and managed **BOMs** and **ECOs** in **Teamcenter** and **SAP-PLM**.
- Architected, built, and validated a **modular fixture** (tooling) controlling a **critical 3 mm interface**; proved **±0.03 mm repeatability** across 10 parts, 3 operators, and 3 cycles.
- Conducted **Monte Carlo tolerance stack-up analyses** across **10+ assemblies**, traced misalignment to the **datum scheme**, and redesigned it to reduce **alignment-related build issues by 25%** versus pre-redesign builds.
- Ran **30+ engineering and design validation (DVT) builds**, defining test criteria and resolving fit and assembly issues with **root-cause analysis** and **DFMEA**.
- Led **supplier DFM/DFA reviews** and **first-article inspections** across **15+ ECOs**, resolving manufacturability issues before release and reducing **supplier lead time by 23%**.
- Generated **$120K in annual savings** by streamlining **Jira-based Agile** design tracking and issue resolution.

University of Minnesota — **Jul 2024 - Jun 2025**
Mechanical Engineer — *Minnesota, USA*

- Engineered a **Stephenson III six-bar linkage** mechanism (MATLAB, LINCAGES) for a **variable rocket nozzle**, varying exit diameter from 0.9 m to 3.3 m; held **transmission angle greater than 30 degrees** and built a **physical prototype**.
- Validated **structural designs** with **ANSYS FEA** (structural, thermal, modal) and **hand calculations**, checking **stress**, **deflection**, **vibration**, and **design margins**.

Emerson / University of Minnesota — **Sep 2023 - May 2024**
Product Development Engineer — *Minnesota, USA*

- Built and tested **11 pre-production units** using **CNC machining**, **welding**, and **3D printing**, then streamlined the original build process to reduce **average build time by 17%**.
- Designed the **enclosure**, **mounting interfaces**, **seals**, and **precision components** in **SolidWorks** with **GD&T** drawings for assembly.
- Championed **DFM/DFA** and **supplier feedback** on **machined** and **injection-molded parts**, reducing **design revisions** from **7 to 3**.

BGR Mining and Infra Limited — **Jun 2020 - Jul 2022**
Mechanical Engineer — *Odisha, India*

- Diagnosed and corrected recurring failures across **20+ production equipment systems** through **root cause analysis** and **corrective action**, reducing **incidents by 15%**.
- Implemented **Lean** and **OEE**-driven process improvements using **Minitab**, reducing **equipment downtime by 6%** and generating approximately **$50K in annual savings**.
- Standardized **preventive-maintenance** and **inspection processes** across production equipment, improving **uptime**, **reliability**, and **equipment service life**.

EDUCATION

University of Minnesota — **Sep 2022 - May 2024**
Master of Science, Mechanical Engineering — *Minnesota, USA*